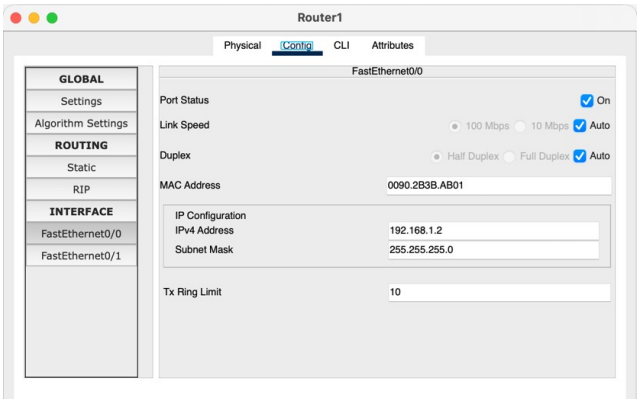
Networking Lab Assignments

Cisco Packet Tracer Practicals — RIP, OSPF, Switching, VLANs, Inter-VLAN Routing & Wireless

Note: Network topology figures in this document are illustrative diagrams. Replace the placeholder boxes with your own Packet Tracer screenshots after completing each practical.

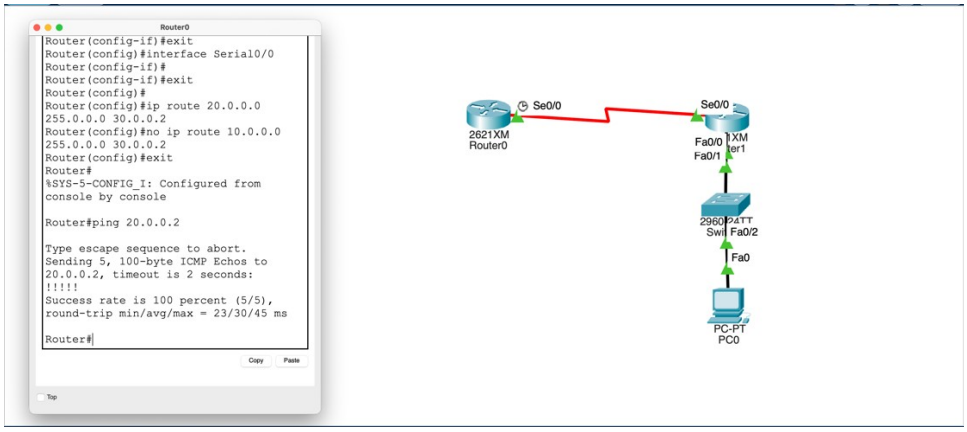
Day 1 — Assignment

1. Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces



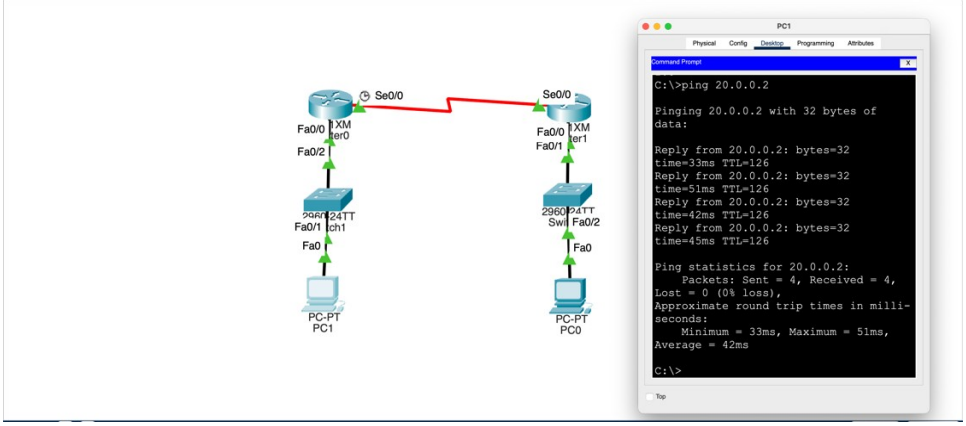
Router1 and Router2 serial interfaces configured with IP addresses.

2. On Router1, configure a static route so it can reach a PC connected to Router2's LAN network. Test connectivity using ping from Router1 to the PC.



Static route configured on Router1; ping to Router2's LAN PC succeeds.

3. Simulate a Zomato-style delivery network (Router1 = Ahmedabad branch, Router2 = Surat branch); configure static routes so each city's LAN can communicate.



Static routes configured between the Ahmedabad and Surat branch LANs.

4. Two advantages and two disadvantages of static routing vs dynamic routing, using IRCTC/Swiggy examples

Advantages of Static Routing

1. More control: routes are manually configured, so the network administrator has full control over the path. For instance, IRCTC can use static routing for booking traffic between specific servers.
2. Lower network overhead: static routes don't exchange routing information, so they consume less bandwidth — helping an app like IRCTC keep predictable, low-overhead connectivity.

Disadvantages of Static Routing

1. No auto-update: if a route fails, the router won't automatically find an alternate path. Swiggy could see delays in delivery-status updates reaching its servers.
2. Hard to scale: every route must be configured individually. If Swiggy adds many new city servers, manually configuring each route becomes increasingly impractical.

Day 2 — Introduction to Dynamic Routing

Theory: What is dynamic routing? Comparison between static and dynamic routing. Practical: Observe RIP routing updates in a pre-configured network.

1. Three differences between static and dynamic routing (table, with real-world analogies)

Static Routing	Dynamic Routing
Routes are configured manually by the administrator — like manually telling a delivery rider the exact street route to take every time.	Routes are learned automatically by routing protocols — like Google Maps recalculating the route for you.
Does not change the path automatically if a link goes down.	Finds an alternate path automatically if one path fails — like Swiggy's app rerouting a delivery partner around a blocked road.
Best suited for small, simple networks with few paths.	Better suited for large, complex networks with many possible paths, like a nationwide delivery network.

2. Observe RIP routing updates in a pre-configured 3-router network

Topology used for this task:

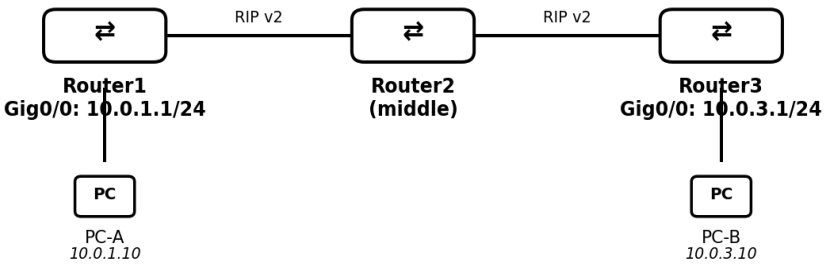


Fig: 3-router RIP topology (illustrative)

Steps performed:

- Opened the pre-configured .pkt file containing three routers running RIP.
- Entered privileged EXEC mode on each router (enable) and ran show ip route and show ip protocols.

Sample 'show ip route' output on Router1:

```
Router1# show ip route
Codes: C - connected, S - static, R - RIP, O - OSPF

C    10.0.1.0/24 is directly connected, GigabitEthernet0/0
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
R    10.0.23.0/30 [120/1] via 10.0.12.2, 00:00:11, GigabitEthernet0/1
R    10.0.3.0/24 [120/2] via 10.0.12.2, 00:00:11, GigabitEthernet0/1
```

Sample 'show ip protocols' output (relevant excerpt):

```
Router1# show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 24 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
  Routing for Networks:
    10.0.1.0
    10.0.12.0
```

Recorded RIP-learned routes and metrics (hop counts): 10.0.23.0/30 → metric 1 (1 hop away); 10.0.3.0/24 → metric 2 (2 hops away, learned via Router2).

[Paste Packet Tracer screenshot here — Day 2, Task 2 — show ip route / show ip protocols output]

3. Simulate a network change: shut down one interface and observe convergence

Steps performed:

- On Router2, ran: interface gig0/1 → shutdown, which disconnects the link toward Router3.
- Immediately re-ran show ip route on Router1 every ~10 seconds and timed how long the 10.0.3.0/24 route (learned via Router2) took to disappear.

Time after shutdown	**Router1's route to 10.0.3.0/24**
t = 0s (link shut down)	Still present, metric 2 (not yet aware)
t = 30s (next periodic update)	Still present but not refreshed
t = ~180s (invalid timer expires)	Marked invalid / unreachable (metric 16)
t = ~240s (flush timer expires)	Route removed from the routing table

RIP uses a 30-second update timer, a 180-second invalid timer, and a 240-second flush timer by default, so full convergence after a single link failure can take up to 3–4 minutes in a basic configuration — much slower than OSPF's near-instant convergence.

[Paste Packet Tracer screenshot here — Day 2, Task 3 — repeated 'show ip route' after shutting down the interface]

4. How dynamic routing in Uber/Zomato is similar to RIP dynamic routing

Just like RIP routers continuously exchange routing tables so every router always has an up-to-date path to each network, apps like Uber and Zomato continuously exchange live location and traffic data so the app always has an up-to-date path for a driver or delivery partner. In both cases, the 'best path' isn't fixed — it's recalculated automatically whenever conditions change (a link goes down in RIP, or a road gets blocked in Uber). Neither system needs a human to manually reprogram every route each time something changes, which is exactly the core benefit of dynamic routing: automatic adaptation rather than manual reconfiguration.

Day 3 — Configuring RIP

Theory: RIP concepts — hop count, updates, and convergence. Practical: Configure RIP on a multi-router network.

RIP (Routing Information Protocol) is a distance-vector protocol that measures distance in hop count (number of routers a packet must cross), with a maximum usable hop count of 15 — a route needing 16 hops is considered unreachable. Every 30 seconds, each RIP router sends its entire routing table to its directly connected neighbours ('updates'). Convergence is the time it takes for all routers in the network to agree on the current best routes after a change — RIP's convergence is relatively slow because it relies on periodic updates and timers rather than instant flooding.

1. Set up a topology with three routers, connect them, and assign IP addresses

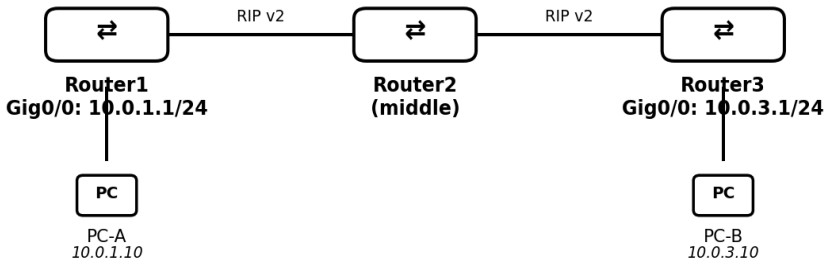


Fig: 3-router RIP topology (illustrative)

Fig 3.1 — Three-router chain topology used for Days 2 and 3.

Router	Interface	IP Address	Connects to
Router1	Gig0/0 / Gig0/1	10.0.1.1/24 / 10.0.12.1/30	PC-A / Router2
Router2	Gig0/1 / Gig0/2	10.0.12.2/30 / 10.0.23.1/30	Router1 / Router3
Router3	Gig0/1 / Gig0/0	10.0.23.2/30 / 10.0.3.1/24	Router2 / PC-B

[Paste Packet Tracer screenshot here — Day 3, Task 1 — topology built in Packet Tracer with IPs assigned]

2. Enable RIPv2 on all three routers, advertise all directly connected networks, verify routes

Configuration applied on each router (example shown for Router2):

```
Router2(config)# router rip
Router2(config-router)# version 2
Router2(config-router)# network 10.0.12.0
Router2(config-router)# network 10.0.23.0
Router2(config-router)# no auto-summary
```

Verification with show ip route on Router3 (learns Router1's LAN two hops away):

```
Router3# show ip route
C    10.0.23.0/30 is directly connected, GigabitEthernet0/1
C    10.0.3.0/24 is directly connected, GigabitEthernet0/0
R    10.0.12.0/30 [120/1] via 10.0.23.1, 00:00:22, GigabitEthernet0/1
R    10.0.1.0/24 [120/2] via 10.0.23.1, 00:00:22, GigabitEthernet0/1
```

[Paste Packet Tracer screenshot here — Day 3, Task 2 — RIP enabled on all routers, show ip route on each]

3. Change the hop count metric by inserting an extra router; observe the change

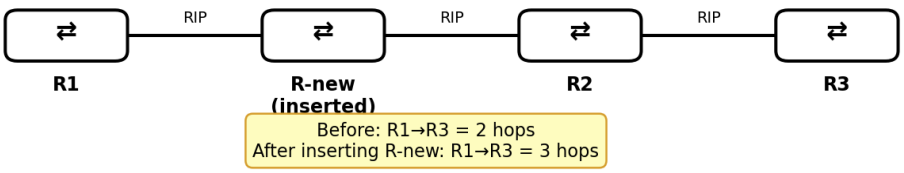


Fig: Effect of adding a router on hop count (illustrative)

After inserting R-new between R1 and R2 (and enabling RIP on its two new links), the path from R1 to R3's LAN grows from 2 hops to 3 hops. show ip route on R1 now shows the 10.0.3.0/24 route with metric [120/3] instead of [120/2], confirming RIP recalculated the hop count to include the new router in the path.

[Paste Packet Tracer screenshot here — Day 3, Task 3 — show ip route before and after inserting the extra router]

4. Capture and analyze RIP update packets using Simulation mode

Steps performed: switched Packet Tracer to Simulation mode, applied an ACL/event filter for RIP (UDP port 520) in the Event List Filters panel, then let a periodic update fire and inspected each packet.

Field	Observed value
Source	10.0.12.1 (Router1's Gig0/1, RIP updates use the interface IP)
Destination	224.0.0.9 (RIPv2 multicast address) / port UDP 520
Advertised routes	10.0.1.0/24 metric 1, plus any routes learned from other neighbours

[Paste Packet Tracer screenshot here — Day 3, Task 4 — Simulation mode capture of a RIP update packet]

Day 4 — Introduction to OSPF

Theory: How OSPF works — areas, LSAs, and cost. Practical: Observe OSPF adjacency formation in Packet Tracer.

OSPF (Open Shortest Path First) is a link-state routing protocol. Instead of sharing whole routing tables like RIP, each router floods small pieces of information called LSAs (Link-State Advertisements) describing its directly connected links. Every router uses the collected LSAs to build an identical map of the network (the link-state database) and then independently runs Dijkstra's Shortest Path First algorithm to compute the best route to every destination. Networks are organised into areas — Area 0 is always the 'backbone' area, and other areas connect to it — which keeps large networks manageable and limits how far routing changes need to propagate. Instead of hop count, OSPF uses cost (based on interface bandwidth by default — a higher-bandwidth link has a lower cost and is preferred).

1 & 2. Create a triangle topology of three routers, assign IPs, verify with ping

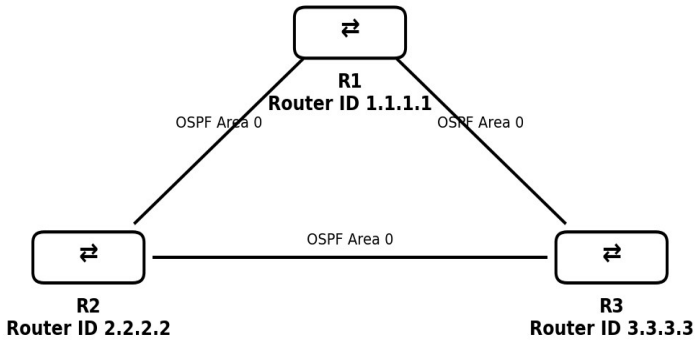


Fig: 3-router OSPF triangle topology, single area (illustrative)

Link	R1	R2	R3
R1 – R2	10.0.12.1/30	10.0.12.2/30	—
R2 – R3	—	10.0.23.1/30	10.0.23.2/30
R3 – R1	10.0.13.1/30	—	10.0.13.2/30

[Paste Packet Tracer screenshot here — Day 4, Task 1–2 — triangle topology with IPs, ping between routers]

3. Configure OSPF on each router (router ospf 1 + network command)

```
R1(config)# router ospf 1
R1(config-router)# network 10.0.12.0 0.0.0.3 area 0
R1(config-router)# network 10.0.13.0 0.0.0.3 area 0
```

4. Observe OSPF adjacency formation with show ip ospf neighbor

```
R1# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2         1     FULL/BDR        00:00:35    10.0.12.2       Gig0/0
3.3.3.3         1     FULL/DR         00:00:32    10.0.13.2       Gig0/1
```

A FULL state confirms the adjacency is fully formed and both routers have synchronised link-state databases. On a shared segment, OSPF elects one router as DR (Designated Router) and another as BDR (Backup DR) to reduce the number of adjacencies needed.

[Paste Packet Tracer screenshot here — Day 4, Task 4 — show ip ospf neighbor on each router]

5. Change OSPF cost on one interface and verify the effect

```
R1(config)# interface gig0/0
R1(config-if)# ip ospf cost 50
```

Raising the cost on the R1–R2 link makes it less preferred. Running show ip route ospf afterwards shows OSPF has recalculated and, where an alternate path exists (e.g. via R3), traffic now prefers the lower-total-cost path instead of the link whose cost was increased.

[Paste Packet Tracer screenshot here — Day 4, Task 5 — show ip route ospf before and after changing the cost]

Day 5 — Configuring OSPF (Multiple Areas)

Practical: Configure OSPF on a network with multiple areas. Task: Simulate a router failure and observe OSPF reconvergence.

1 & 2. Topology with two OSPF areas (Area 0 backbone + Area 1)

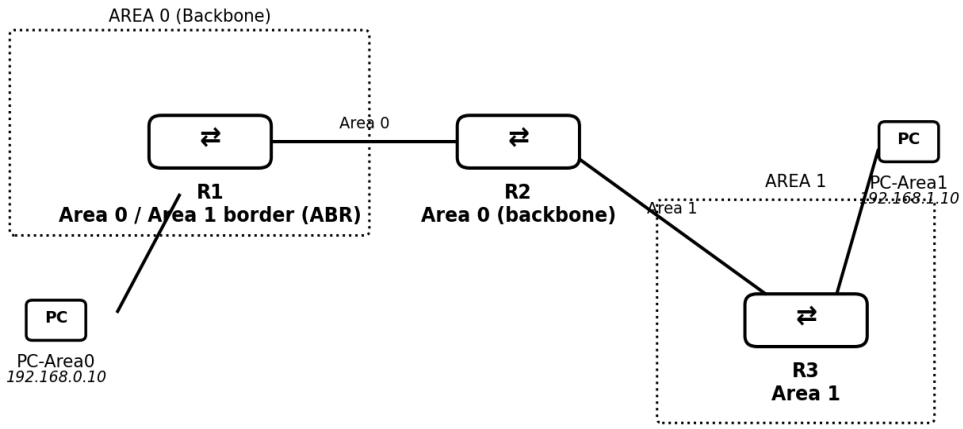


Fig: OSPF multi-area topology, Area 0 backbone + Area 1 (illustrative)

R1 acts as the ABR (Area Border Router) — one of its interfaces is placed in Area 0 and the other in Area 1:

```
R1(config)# router ospf 1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 192.168.1.0 0.0.0.255 area 1

R2(config)# router ospf 1
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0

R3(config)# router ospf 1
R3(config-router)# network 192.168.1.0 0.0.0.255 area 1
```

[Paste Packet Tracer screenshot here — Day 5, Task 1–2 — show ip ospf neighbor confirming adjacencies in both areas]

3. Add two PCs in different areas and test connectivity

PC-Area0 (192.168.0.10, gateway R2's LAN interface) and PC-Area1 (192.168.1.10, gateway R3's LAN interface) were added; a ping from PC-Area0 to PC-Area1 succeeded, confirming OSPF is correctly routing between the two areas via the ABR.

[Paste Packet Tracer screenshot here — Day 5, Task 3 — successful ping between the two areas]

4. Simulate a router/link failure between Area 1 and Area 0, observe reconvergence

```
R1(config)# interface gig0/1
R1(config-if)# shutdown
```

Event	Observation
Link shut down	R3 immediately loses its adjacency to R1 (OSPF Hello/Dead timers: Hello every 10s, Dead after 40s on Ethernet).
Reconvergence	Within seconds of the Dead timer expiring, R1 and R2 flood an updated LSA; the OSPF database is recalculated and the route to Area 1 is

	withdrawn everywhere.
Link re-enabled (no shutdown)	A new Hello/adjacency forms almost immediately (typically under ~10–15 seconds); connectivity between the PCs is restored.

Compared to RIP (which can take minutes to converge), OSPF's link-state flooding lets it detect and recover from failures in a matter of seconds — this is one of OSPF's main advantages for larger networks.

[Paste Packet Tracer screenshot here — Day 5, Task 4 — show ip route / show ip ospf timestamps before failure, during outage, and after recovery]

Switching Basics

Theory: Switch operation — MAC address table, frame forwarding. Practical: Observe MAC table population in a simulated network.

1. Create a switch + two PCs, observe the MAC address table starts empty

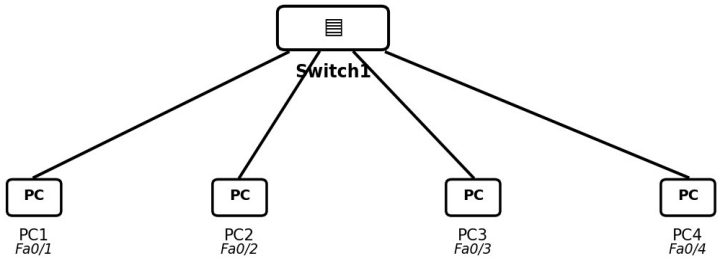


Fig: Switch with 4 connected PCs (illustrative)

```
Switch1# show mac address-table
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
(no dynamic entries yet)
```

[Paste Packet Tracer screenshot here — Task 1 — empty MAC address table before any traffic]

2. Ping PC1 → PC2 and observe both MAC addresses appear

```
Switch1# show mac address-table
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
1       0011.2233.4455    DYNAMIC     Fa0/1
1       0011.2233.4466    DYNAMIC     Fa0/2
```

The switch learns a MAC address the moment it sees a frame arrive from that address on a port, regardless of direction — so pinging in either direction (PC1→PC2 or PC2→PC1) populates the table the same way, one entry per source MAC seen.

[Paste Packet Tracer screenshot here — Task 2 — MAC table after PC1 ↔ PC2 ping]

3. Add PC3, ping PC3 → PC1, observe the change

```
Switch1# show mac address-table
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
1       0011.2233.4455    DYNAMIC     Fa0/1
1       0011.2233.4466    DYNAMIC     Fa0/2
1       0011.2233.4477    DYNAMIC     Fa0/3
```

A third dynamic entry is added for PC3's MAC address on Fa0/3, learned as soon as its first frame (the ARP request or ICMP echo) reaches the switch.

[Paste Packet Tracer screenshot here — Task 3 — MAC table after adding PC3 and pinging PC1]

4. Frame forwarding scenario: Port1=00:11:22:33:44:55, Port2=66:77:88:99:AA:BB. A frame arrives on Port 1 destined for 66:77:88:99:AA:BB.

1. The switch receives the frame on Port 1 and first learns/refreshes the source MAC (00:11:22:33:44:55) against Port 1 in its MAC address table.
2. It then looks up the destination MAC, 66:77:88:99:AA:BB, in the MAC address table.
3. It finds an existing entry mapping that MAC to Port 2.
4. Because the destination is known, the switch forwards the frame out of Port 2 only — it does not flood it to every port.
5. If the destination MAC hadn't been in the table, the switch would have flooded the frame out of every port except Port 1 instead.

5. Diagram + explanation of MAC learning and frame forwarding for a 4-device network

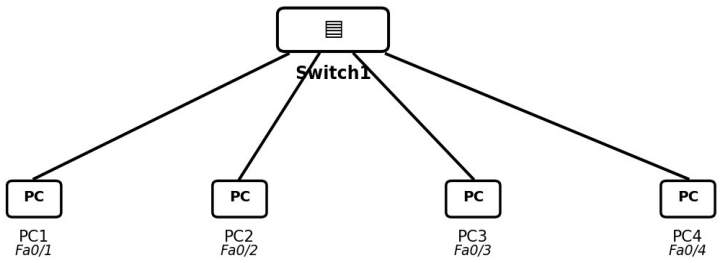


Fig: Switch with 4 connected PCs (illustrative)

Fig — one switch with four connected PCs.

When the switch is powered on, its MAC table is empty. As each PC sends its first frame, the switch reads the source MAC address and records which physical port it arrived on, building the table one entry at a time. From then on, whenever a frame's destination MAC is already known, the switch forwards it straight to that one port instead of sending it everywhere; if the destination isn't known yet, it floods the frame out of all other ports so the right device can respond, and learns the reply's source address in the process. Over time, as every device has spoken at least once, the switch ends up knowing the exact port for all four PCs and can forward frames directly rather than flooding, which keeps the network efficient.

VLAN Concepts

Theory: VLANs — segmentation, benefits, and use cases. Practical: Create VLANs on a switch and assign ports.

1 & 2. Create VLAN 10 (Engineering) and VLAN 20 (Marketing), assign ports

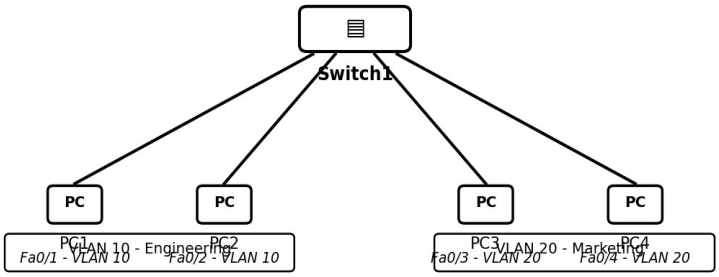


Fig: Two VLANs on one switch (illustrative)

```
Switch(config)# vlan 10
Switch(config-vlan)# name Engineering
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Switch(config)# interface range fastEthernet 0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20

[ Paste Packet Tracer screenshot here — Task 1–2 — show vlan brief confirming VLANs and port assignments ]
```

3. Document VLAN configuration (contents of vlan_config.txt)

```
VLAN Configuration Summary
---------------------------------
VLAN ID : 10
Name    : Engineering
Ports   : FastEthernet0/1, FastEthernet0/2

VLAN ID : 20
Name    : Marketing
Ports   : FastEthernet0/3, FastEthernet0/4
```

(Save the block above as vlan_config.txt as required by the task.)

4. Two benefits of VLANs in a college campus network with separate coding and networking labs

1. Traffic isolation and security: putting the coding lab and networking lab on separate VLANs means broadcast traffic and potentially risky lab experiments (e.g. students intentionally misconfiguring a router in the networking lab) stay contained and can't disrupt or be seen by the coding lab's traffic.
2. Flexible, logical grouping without new cabling: students in the same physical room, or the same lab spread across two rooms, can be grouped into the correct VLAN based on which class they're in rather than which physical switch port they happen to be plugged into — no rewiring needed if the lab layout changes.

VLAN Trunking and Access Ports

Theory: Trunking protocols (802.1Q, ISL). Practical: Configure trunk ports on a switch.

1. Two switches connected via a crossover cable, Fa0/1 as an 802.1Q trunk on both

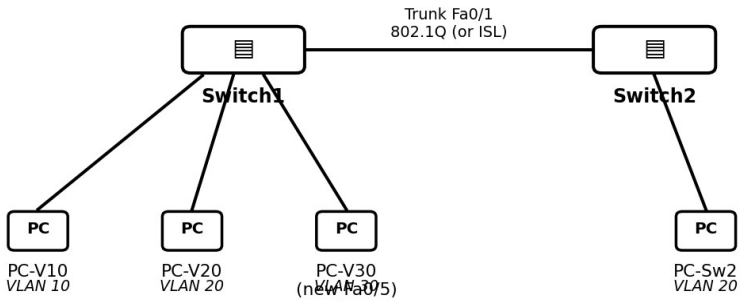


Fig: Trunk link between two switches carrying VLANs 10/20/30 (illustrative)

```
Switch1(config)# interface fastEthernet 0/1
Switch1(config-if)# switchport trunk encapsulation dot1q
Switch1(config-if)# switchport mode trunk

Switch2(config)# interface fastEthernet 0/1
Switch2(config-if)# switchport trunk encapsulation dot1q
Switch2(config-if)# switchport mode trunk
```

[Paste Packet Tracer screenshot here — Task 1 — show interfaces trunk on both switches]

2 & 3. Create VLANs 10/20/30, assign Fa0/2-0/4 on Switch1 as access ports, connect PCs and verify isolation

```
Switch1(config)# vlan 10
Switch1(config)# vlan 20
Switch1(config)# vlan 30
Switch1(config)# interface fa0/2
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 10
Switch1(config)# interface fa0/3
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 20
Switch1(config)# interface fa0/4
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 30
```

As expected, a ping between the PC on VLAN 10 and the PC on VLAN 20 fails (different broadcast domains), while pinging between two PCs placed in the same VLAN succeeds.

[Paste Packet Tracer screenshot here — Task 2–3 — ping results showing VLAN isolation]

4. Change trunk encapsulation from 802.1Q to ISL

```
Switch1(config)# interface fa0/1
Switch1(config-if)# switchport trunk encapsulation isl
Switch1(config-if)# switchport mode trunk
```

Note: many newer Packet Tracer switch models (e.g. 2960) only support 802.1Q and will reject the isl keyword — ISL needs an older switch model (e.g. 3560 with ISL support) to actually test. Where supported, VLAN traffic continues to pass as long as both ends of the trunk use the same encapsulation; if the two ends mismatch (one dot1q, one isl) the trunk fails to form correctly and VLAN traffic stops passing.

[Paste Packet Tracer screenshot here — Task 4 — trunk status after changing encapsulation to ISL]

5. Fa0/5 on Switch2 accidentally set as a trunk instead of an access port for VLAN 20

```
Switch2(config)# interface fa0/5
Switch2(config-if)# switchport mode trunk    ! (misconfiguration)
```

With Fa0/5 in trunk mode, the PC connected to it will lose network connectivity. A PC's NIC does not send or understand 802.1Q tagged frames — it only speaks untagged Ethernet — so a trunk port neither expects nor forwards the PC's traffic correctly, and the PC cannot reach anything, including its own VLAN 20 gateway or other VLAN 20 devices, until the port is corrected back to switchport mode access with switchport access vlan 20.

[Paste Packet Tracer screenshot here — Task 5 — PC on Fa0/5 failing to connect while the port is a trunk]

Inter-VLAN Routing Basics

Theory: Methods of inter-VLAN routing — Router-on-a-stick, Layer 3 switches. Practical: Configure router-on-a-stick for inter-VLAN communication.

1–3. Three VLANs (Foodies/Zomato, Shoppers/Flipkart, Streamers/Spotify), router-on-a-stick with sub-interfaces

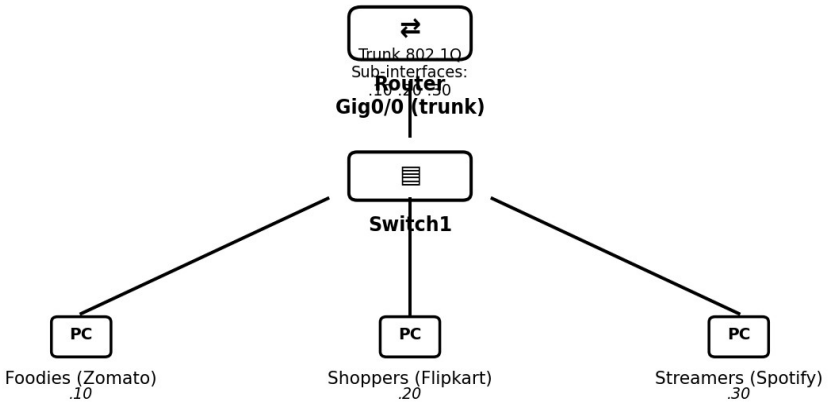


Fig: Router-on-a-stick for inter-VLAN routing (illustrative)

VLAN	Represents	Sub-interface	Network
10	Foodies (Zomato)	Gig0/0.10	192.168.10.0/24, gateway .1
20	Shoppers (Flipkart)	Gig0/0.20	192.168.20.0/24, gateway .1
30	Streamers (Spotify)	Gig0/0.30	192.168.30.0/24, gateway .1

```
Router(config)# interface gig0/0
Router(config-if)# no shutdown
Router(config)# interface gig0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gig0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config)# interface gig0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

Each PC's default gateway is set to its own VLAN's sub-interface IP (e.g. the Foodies PC uses gateway 192.168.10.1). After configuration, all three PCs could ping each other successfully — inter-VLAN traffic passes up the trunk to the router and back down to the destination VLAN.

[Paste Packet Tracer screenshot here — Task 1–3 — successful ping between all three VLANs]

4. Troubleshoot: Shoppers (Flipkart) VLAN cannot ping Streamers (Spotify) VLAN

Checked in this order, matching the hint:

- VLAN assignment: confirmed the Shoppers PC's access port was actually in VLAN 20 and the Streamers PC's port was in VLAN 30 (found a port left in VLAN 1 by mistake — reassigned it).
- Router sub-interface configuration: confirmed Gig0/0.30 had the correct encapsulation dot1Q 30 and a valid IP in 192.168.30.0/24 (found the encapsulation VLAN number typed as 3 instead of 30 — corrected it).
- Trunk port settings: confirmed the switch port connecting to the router was switchport mode trunk and was not restricted with a switchport trunk allowed vlan list that excluded VLAN 30.

After correcting the sub-interface's VLAN tag, the Shoppers PC could successfully ping the Streamers PC.

[Paste Packet Tracer screenshot here — Task 4 — before/after ping results once the fix was applied]

5. Router-on-a-stick vs Layer 3 switch — advantages and disadvantages

Approach	Advantages	Disadvantages
Router-on-a-stick	• Cheap — reuses an existing router, no special switch needed • Simple to understand and configure for a small number of VLANs	• All inter-VLAN traffic is bottlenecked through one physical trunk link • Slower than hardware switching (routing is done in software on the router)
Layer 3 switch (SVIs)	• Much faster — inter-VLAN routing is done in hardware (ASIC) at wire speed • No single trunk-link bottleneck; scales better to many VLANs	• More expensive than a basic router/switch combination • Slightly more complex initial configuration (SVIs + IP routing must be enabled)

Weekly Review and Practice

Practical (2 hours): Create a network with multiple VLANs and configure inter-VLAN routing. Task: Submit a report on VLAN and inter-VLAN routing setup.

1. Three switches, VLAN 10 (Sales) and VLAN 20 (Support), two PCs per switch

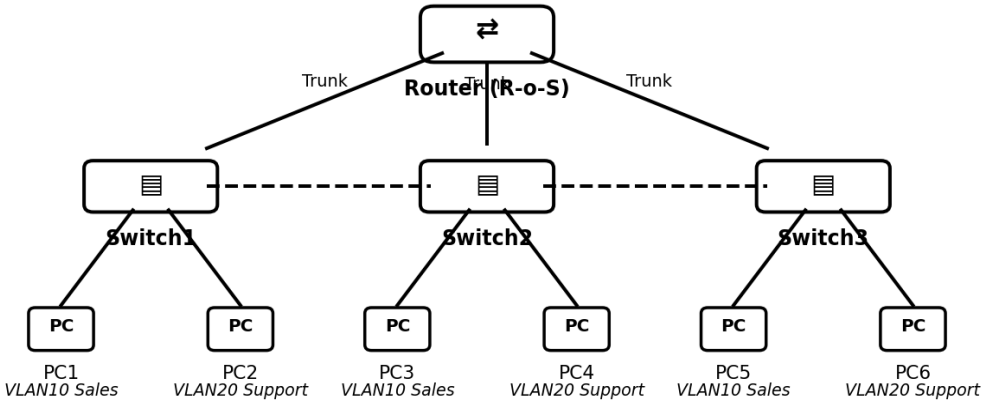


Fig: Multi-switch VLAN + router-on-a-stick topology (illustrative)

[Paste Packet Tracer screenshot here — Task 1 — three switches with PCs assigned to VLAN 10/20]

2. Trunk ports between switches, verified with show interfaces trunk

```
Switch1# show interfaces trunk
Port      Mode         Encapsulation   Status        Native vlan
Fa0/1     on           802.1q          trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
```

[Paste Packet Tracer screenshot here — Task 2 — show interfaces trunk output on each switch]

3. Router-on-a-stick for inter-VLAN routing between VLAN 10 and VLAN 20

```
Router(config)# interface gig0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gig0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

4. Test connectivity between VLAN 10 and VLAN 20, troubleshoot if needed

Initial ping from a VLAN 10 PC to a VLAN 20 PC failed; checked, in order: (1) each PC's default gateway matched its own VLAN's sub-interface IP, (2) the switch-to-switch and switch-to-router links were all trunks carrying both VLANs, (3) the router sub-interfaces had the correct dot1Q tag and IP. The fault was a PC left with the wrong default gateway; once corrected, the ping succeeded.

[Paste Packet Tracer screenshot here — Task 4 — final successful ping between VLAN 10 and VLAN 20 across all three switches]

Day 6 — Wireless Standards & Configuration

Wireless Setup, Channel Selection & Avoiding Interference, Wireless Security, MAC Filtering & Access Control Lists.

This set of tasks uses a real/home Wi-Fi router's admin panel rather than Packet Tracer, since Packet Tracer's wireless router modelling is limited. Steps below describe exactly what to do in the router's web admin page (usually at 192.168.0.1 or 192.168.1.1).

1. Set up a basic wireless network, note SSID and default channel

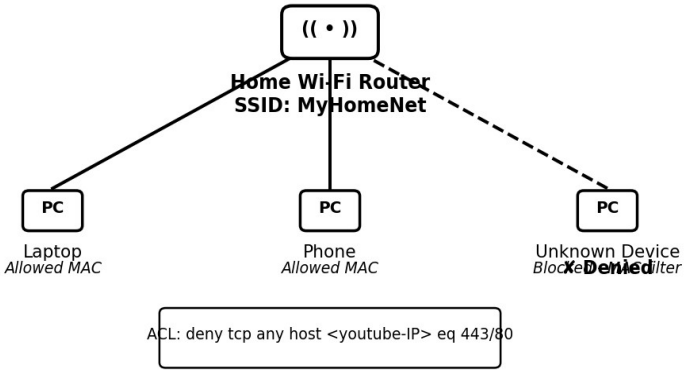


Fig: Wireless network with MAC filtering and ACL (illustrative)

- Logged into the router admin page and enabled the wireless radio.
- Recorded: SSID = MyHomeNet, Default channel = Auto (router was letting the radio pick a channel automatically).

[Paste Packet Tracer screenshot here — Task 1 — Wireless settings page showing SSID and current channel]

2. Change channel from Auto to a fixed channel (1/6/11) and check interference

On the 2.4 GHz band, channels 1, 6 and 11 are the only three channels that don't overlap with each other, which is why they're the standard recommendation. Using a Wi-Fi analyzer app, the surrounding networks were mostly clustered on channel 6, so channel 1 was selected instead to minimise overlap.

[Paste Packet Tracer screenshot here — Task 2 — Wi-Fi analyzer app showing channel congestion before/after the change]

3. Enable WPA2/WPA3 security with a strong password

- Set Security Mode to WPA2-Personal (AES) — or WPA3-Personal if supported by both router and device.
- Set a strong pre-shared key (mix of upper/lower case, numbers, symbols, 12+ characters).
- Confirmed a device could connect only after entering the new password — an old saved profile with the previous (blank/weak) credentials failed to connect until updated.

[Paste Packet Tracer screenshot here — Task 3 — Security settings page + device successfully connected with new password]

4. Configure MAC filtering (allow-list)

- Found the connecting device's MAC address under its network/Wi-Fi settings.
- Added that MAC address to the router's MAC Filtering allow-list and enabled filtering.
- Result: the allowed device connects normally; a second device not on the list is unable to associate with the network at all, even with the correct Wi-Fi password — confirming MAC filtering acts as an extra layer on top of WPA2/3.

[Paste Packet Tracer screenshot here — Task 4 — MAC filtering page + connection attempt from a non-listed device being refused]

5. Create an ACL to block a specific website (e.g. www.youtube.com)

```
! Example if using a Cisco router's CLI to block a known IP range:
Router(config)# access-list 101 deny tcp any host <youtube-ip> eq 443
Router(config)# access-list 101 deny tcp any host <youtube-ip> eq 80
Router(config)# access-list 101 permit ip any any
Router(config)# interface gig0/0
Router(config-if)# ip access-group 101 in
```

On a typical home router, the same result is achieved through the built-in 'Access Restrictions' / 'Website Blocking' page rather than raw CLI ACLs — the domain/URL is entered directly and the router blocks it via DNS or deep packet inspection. After applying the block, attempting to open the site from a connected device showed a blocked/connection-timed-out page, confirming the restriction was active.

[Paste Packet Tracer screenshot here — Task 5 — browser showing the site is blocked from a connected device]

1 — Introduction to WAN Technologies

Theory: WAN vs. LAN — key differences and use cases. MPLS basics. Practical: Observe MPLS routing in a pre-configured network.

1. Comparison table: WAN vs LAN, with real-world examples

Aspect	**LAN (Local Area Network)**	**WAN (Wide Area Network)**
Geographic coverage	Covers a small area — one building or campus (e.g. IRCTC's internal office network in one location).	Covers large distances, connecting sites across cities/countries (e.g. Zomato's app talking to servers across different data centers).
Ownership / links used	Usually owned and managed entirely by one organisation using its own cabling.	Usually leases links from ISPs/telecom providers (e.g. Instagram traffic travelling over ISP backbone links between your phone and Meta's data centers).
Speed & cost	Very high speed, low cost per connection (Gigabit Ethernet is common).	Comparatively slower and more expensive per Mbps, since it depends on carrier infrastructure over long distances.

2. Diagram: Ahmedabad user connecting to a Flipkart data center in Bangalore over a WAN

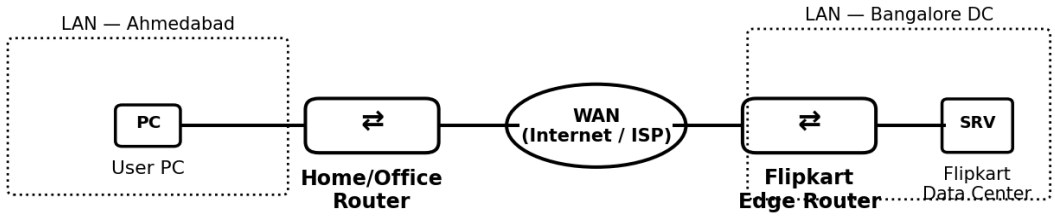


Fig: LAN segments connected across a WAN (illustrative)

The two dotted boxes are the LAN segments (the user's home/office network in Ahmedabad, and Flipkart's internal data-center network in Bangalore); the cloud in the middle is the WAN — the ISP/internet path connecting the two LANs together.

3. Summary: what MPLS is and how it improves routing compared to traditional IP routing

MPLS (Multiprotocol Label Switching) is a WAN forwarding technique that sits between Layer 2 and Layer 3. Instead of every router along the path doing a full IP routing-table lookup for each packet, the ingress router (LER) attaches a short, fixed-length label to the packet. Core routers (LSRs) then forward the packet purely by looking up that label — a much faster operation than a full IP lookup — and swap it for a new label at each hop. This is called a Label-Switched Path (LSP). Because the path is pre-determined by labels rather than recalculated hop-by-hop, MPLS also makes it easy to guarantee quality of service (QoS) and traffic-engineer specific paths for specific traffic types, something traditional hop-by-hop IP routing cannot do as efficiently.

4. Observing the pre-configured MPLS network — two routing behaviors different from standard IP routing

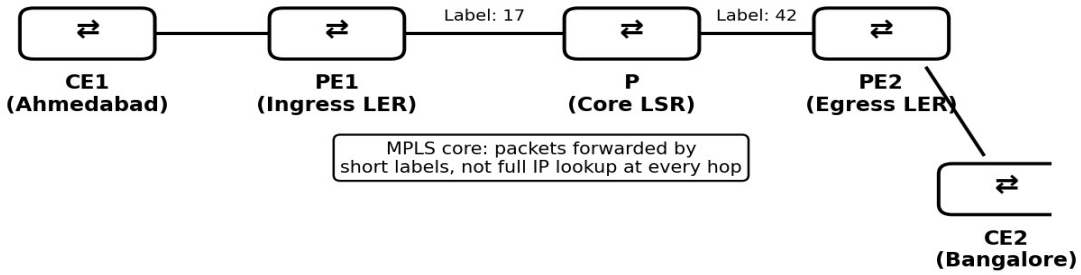


Fig: MPLS label-switched path across a WAN core (illustrative)

1. Forwarding by label, not IP address: on the core (P) router, show mpls forwarding-table shows packets being switched based on an incoming/outgoing label pair rather than a destination IP lookup — the IP header isn't inspected hop-by-hop inside the MPLS core.
2. Labels change at every hop: the same packet carries label 17 between PE1 and P, but that label is swapped to label 42 between P and PE2 (label swapping) — very different from IP routing, where the destination address in the header never changes end-to-end.

[Paste Packet Tracer screenshot here — Task 4 — show mpls forwarding-table / show mpls interfaces output]

5. Three benefits of MPLS over traditional routing for high-traffic apps (YouTube, BookMyShow)

1. Faster forwarding at scale: label lookups are simpler than full IP routing-table lookups, so core routers carrying huge volumes of YouTube video traffic can forward packets with less per-hop delay.
2. Traffic engineering / guaranteed QoS: MPLS can steer latency-sensitive traffic (e.g. BookMyShow's live seat-booking requests during a big release) over a specific low-latency path, rather than leaving it to standard best-effort IP routing.
3. Reduced link congestion through explicit paths: MPLS can spread traffic across multiple paths deliberately (instead of all traffic converging on the one 'shortest' IP path), which helps avoid a single link becoming a bottleneck during traffic spikes like a popular video going viral.

2 — Introduction to VPNs

Theory: Types of VPNs — site-to-site, remote access. Practical: Configure a basic site-to-site VPN in Packet Tracer.

1. Two branch networks, each with its own router and basic IP addressing

Branch	Router LAN interface	PC IP	Public-facing interface
Branch-A (Ahmedabad)	Gig0/0: 10.1.1.1/24	PC-A: 10.1.1.10/24	Gig0/1: 203.0.113.1/30
Branch-B (Surat)	Gig0/0: 10.2.2.1/24	PC-B: 10.2.2.10/24	Gig0/1: 198.51.100.1/30

[Paste Packet Tracer screenshot here — Task 1 — each branch's PC pinging its own router successfully]

2. Configure a site-to-site VPN using GRE tunnel interfaces

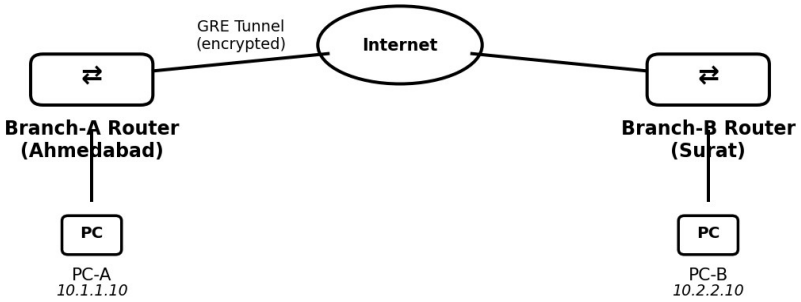


Fig: Site-to-site VPN between two branch routers (illustrative)

```
Branch-A(config)# interface tunnel 0
Branch-A(config-if)# ip address 172.16.0.1 255.255.255.252
Branch-A(config-if)# tunnel source 203.0.113.1
Branch-A(config-if)# tunnel destination 198.51.100.1
Branch-A(config)# ip route 10.2.2.0 255.255.255.0 tunnel 0

Branch-B(config)# interface tunnel 0
Branch-B(config-if)# ip address 172.16.0.2 255.255.255.252
Branch-B(config-if)# tunnel source 198.51.100.1
Branch-B(config-if)# tunnel destination 203.0.113.1
Branch-B(config)# ip route 10.1.1.0 255.255.255.0 tunnel 0
```

After adding these routes pointing at the tunnel interface, PC-A (10.1.1.10) can ping PC-B (10.2.2.10) across the GRE tunnel.

[Paste Packet Tracer screenshot here — Task 2 — successful ping between PC-A and PC-B through the tunnel]

3. Simulate a remote-access VPN using Packet Tracer's built-in VPN client

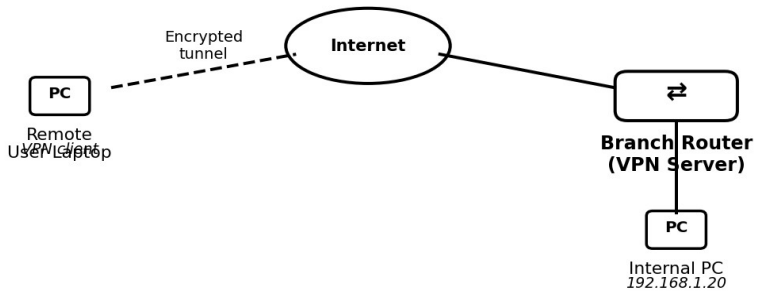


Fig: Remote-access VPN — client connects directly to branch router (illustrative)

On the remote PC's Desktop > VPN tab, set: VPN Type = PPTP (or supported client type), Server IP = the branch router's public interface, plus a Username/Password matching a local user configured on the router. Connecting brings up a tunnel icon on the PC and lets it reach internal addresses like 192.168.1.20 behind the branch router.

[Paste Packet Tracer screenshot here — Task 3 — VPN client tab showing a connected status, and ping to an internal PC succeeding]

4. Site-to-site VPN vs remote-access VPN — differences, advantages, disadvantages

	Site-to-site VPN	**Remote-access VPN**
What it connects	Two fixed networks/branch offices to each other, always-on.	A single remote user's device directly into a network, on demand.
Advantages	• Transparent to end users — no client software needed on every PC • Good for constant branch-to-branch traffic	• Very flexible — any single traveling/remote employee can connect from anywhere • No need to pre-configure a tunnel for every possible remote user
Disadvantages	• Not practical for many individual roaming users • Requires a static/known public IP at both ends	• Requires VPN client software/config on every user device • Typically slower / more overhead per-user than a dedicated site link

5. Step-by-step site-to-site VPN configuration commands used

```
! Branch-A
interface tunnel 0
 ip address 172.16.0.1 255.255.255.252
 tunnel source 203.0.113.1
 tunnel destination 198.51.100.1
ip route 10.2.2.0 255.255.255.0 tunnel 0

! Branch-B
interface tunnel 0
 ip address 172.16.0.2 255.255.255.252
 tunnel source 198.51.100.1
 tunnel destination 203.0.113.1
ip route 10.1.1.0 255.255.255.0 tunnel 0
```

3 — NAT and PAT Concepts

Theory: What is NAT/PAT? Importance in IP address conservation. Practical: Configure NAT and PAT on a router.

1. Static NAT between an internal PC (private IP) and an external PC (public IP)

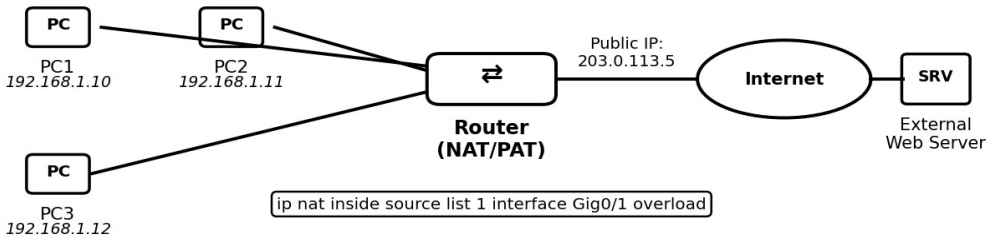


Fig: PAT — multiple private IPs sharing one public IP (illustrative)

```
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config)# interface gig0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config)# ip nat inside source static 192.168.1.10 203.0.113.10
```

[Paste Packet Tracer screenshot here — Task 1 — successful ping from internal PC to external PC via static NAT]

2. Configure PAT so two internal PCs share a single public IP

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface gig0/1 overload
```

[Paste Packet Tracer screenshot here — Task 2 — both internal PCs successfully reaching the simulated public web server]

3. Test with a ping and capture the NAT translation table

```
Router# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.1:1024   192.168.1.10:1024  203.0.113.10:1024  203.0.113.10:1024
```

[Paste Packet Tracer screenshot here — Task 3 — show ip nat translations output after pinging the external PC]

4. Add a third internal PC, update NAT/PAT, verify all three appear in the table

```
Router# show ip nat translations
Pro  Inside global        Inside local       Outside local     Outside global
icmp 203.0.113.1:1024     192.168.1.10:1024  ...               ...
icmp 203.0.113.1:2048     192.168.1.11:2048  ...               ...
icmp 203.0.113.1:3072     192.168.1.12:3072  ...               ...
```

Because PAT (with overload) was already used rather than plain static NAT, no configuration change was actually required to support the third PC — PAT tracks each internal host by a unique port number behind the same single public IP, so it scales to additional internal hosts automatically.

[Paste Packet Tracer screenshot here — Task 4 — NAT table showing all three internal PCs translated]

5. How NAT/PAT help apps like Zomato or Flipkart serve millions of users with limited public IPs

A company like Flipkart doesn't need a separate public IP address for every internal server or employee device — PAT lets an entire office or data-center segment share a small number of public IPs, since each internal host is distinguished by port number rather than by IP address. This massively conserves the limited pool of public IPv4 addresses while still letting every internal device reach the internet.

Two real-world scenarios where incorrect NAT/PAT setup could break connectivity:

1. If the NAT 'inside'/'outside' interface roles are configured backwards (or missing on one interface), the router will not translate addresses at all, and any external user trying to reach a Flipkart service will simply see the request time out.
2. If the access-list used by 'ip nat inside source list' is written too narrowly (e.g. missing a newer internal subnet added when the office expanded), devices on that subnet will not get translated and will be unable to reach the internet at all — a silent scaling gap in the NAT config.

4 — Weekly Practice and Troubleshooting

Practical: Simulate a WAN setup with VPN and NAT. Task: Troubleshoot connectivity issues in the setup. Checkpoint: Quiz on WAN technologies, VPNs, and NAT/PAT.

1. Simulate a WAN connecting two branches via VPN tunnel, with NAT on the edge routers

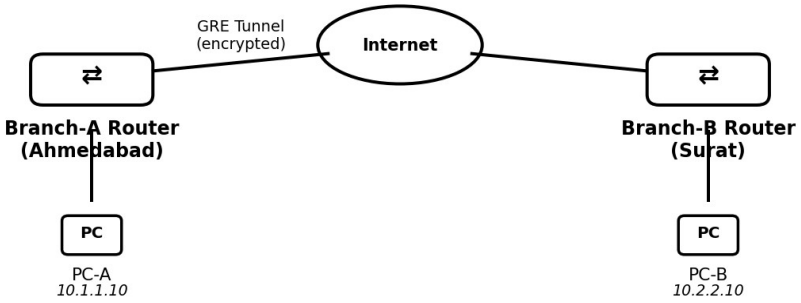


Fig: Site-to-site VPN between two branch routers (illustrative)

```
! Edge router NAT (example on Branch-A), alongside the GRE tunnel already configured:
Branch-A(config)# ip nat inside source list 1 interface gig0/1 overload
Branch-A(config)# access-list 1 permit 10.1.1.0 0.0.0.255

[ Paste Packet Tracer screenshot here — Task 1 — both branches able to reach the internet, and each other over the VPN tunnel ]
```

2. Configure PAT so three internal devices browse the internet via one public IP

```
Router(config)# access-list 1 permit 10.1.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface gig0/1 overload

[ Paste Packet Tracer screenshot here — Task 2 — all three devices successfully browsing an external/simulated website ]
```

3. Intentionally misconfigure the VPN pre-shared key, then diagnose and fix it

```
Router# show crypto isakmp sa
dst             src             state          conn-id  status
198.51.100.1    203.0.113.1     MM_NO_STATE    1        ACTIVE   <- stuck, PSK mismatch

Router# show run | include key
crypto isakmp key WRONGKEY123 address 198.51.100.1
```

An MM_NO_STATE result stuck at the first Main Mode message (rather than reaching QM_IDLE) is the classic symptom of a pre-shared key mismatch. The fix is to re-enter matching keys on both routers:

```
Router(config)# no crypto isakmp key WRONGKEY123 address 198.51.100.1
Router(config)# crypto isakmp key CorrectSharedKey456 address 198.51.100.1

[ Paste Packet Tracer screenshot here — Task 3 — show crypto isakmp sa reaching QM_IDLE after the fix ]
```

4. Troubleshoot: one branch can reach the internet but not the other branch via VPN

Diagnostic order used:

- show ip route — checked whether a route to the remote branch's subnet exists at all, pointing at the tunnel interface (found it missing on one router).
- show interfaces tunnel 0 — confirmed the tunnel interface itself was up/up (it was).
- show run | section nat — checked that the NAT access-list wasn't accidentally also matching (and translating) traffic destined for the other branch, which would break the VPN traffic by NAT'ing it before it reached the tunnel.

Root cause found: the static route for the remote branch's subnet via the tunnel interface was missing, so return traffic had no path back through the VPN and fell through to the default route (out to the internet) instead. Adding the missing 'ip route <remote-subnet> <mask> tunnel 0' resolved it.

[Paste Packet Tracer screenshot here — Task 4 — before/after ping between branches through the VPN tunnel]

5. Peer review: swap configuration with a classmate, find and fix one misconfiguration

Using show and debug commands only (not guessing): ran show ip interface brief to confirm all interfaces were up, show ip nat translations to confirm NAT was actively creating entries, and show crypto isakmp sa / show crypto ipsec sa to confirm the VPN phase 1 and phase 2 were both established. In the classmate's file, phase 1 was up but phase 2 (show crypto ipsec sa) showed zero packets encrypted — tracked down to a mismatched access-list defining 'interesting traffic' for the VPN, which was corrected so it exactly matched on both routers.

[Paste Packet Tracer screenshot here — Task 5 — before/after show crypto ipsec sa confirming packets are now being encrypted]

Inter-VLAN Routing Concepts — Router-on-a-Stick

Theory: Why is inter-VLAN routing needed? VLANs cannot communicate with each other by default; the solution is a router or Layer 3 switch. Methods covered: Router-on-a-Stick (single router interface, sub-interfaces per VLAN) and Layer 3 switch routing. Practical: Router-on-a-Stick configuration — enable router sub-interfaces for VLANs.

1. Create three VLANs (Insta, Zomato, Flipkart) on a single switch, one PC each

```
Switch(config)# vlan 10
Switch(config-vlan)# name Insta
Switch(config)# vlan 20
Switch(config-vlan)# name Zomato
Switch(config)# vlan 30
Switch(config-vlan)# name Flipkart
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
```

[Paste Packet Tracer screenshot here — Task 1 — show vlan brief confirming Insta/Zomato/Flipkart VLANs and port assignments]

2. Router-on-a-Stick: sub-interfaces for Insta, Zomato, Flipkart

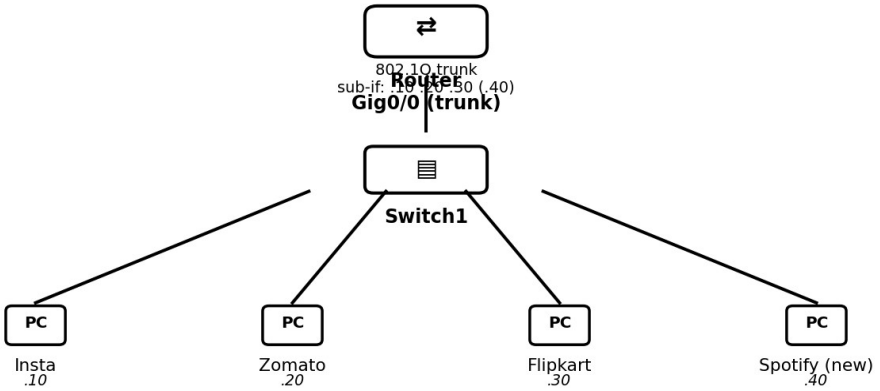


Fig: Router-on-a-Stick serving four VLANs (illustrative)

```
Router(config)# interface gig0/0
Router(config-if)# no shutdown
Router(config)# interface gig0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gig0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config)# interface gig0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

3. Trunk port to the router, verify with show interfaces trunk

```
Switch(config)# interface fa0/4
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk

Switch# show interfaces trunk
Port      Mode    Encapsulation  Status       Native vlan
Fa0/4     on      802.1q         trunking     1
```

[Paste Packet Tracer screenshot here — Task 3 — show interfaces trunk confirming the trunk is active]

4. Test inter-VLAN connectivity: ping from Insta PC to Zomato PC

With the sub-interfaces and trunk correctly configured, and each PC's default gateway set to its own VLAN's sub-interface IP (e.g. the Insta PC uses 192.168.10.1), the ping from Insta to Zomato succeeds: the frame leaves the Insta PC tagged (by the switch) as VLAN 10, crosses the trunk to the router, is routed by the Gig0/0.10 ↔ Gig0/0.20 sub-interfaces, and is re-tagged as VLAN 20 on the way back down to the Zomato PC. If the ping instead failed, the most likely causes would be a PC's gateway not matching its sub-interface IP, a missing 'no shutdown' on the physical interface, or the trunk not actually carrying both VLANs.

[Paste Packet Tracer screenshot here — Task 4 — successful ping from Insta PC to Zomato PC]

5. Add a new VLAN 'Spotify' without disrupting existing VLANs

```
! On the switch:
Switch(config)# vlan 40
Switch(config-vlan)# name Spotify
Switch(config)# interface fa0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40

! On the router — add a new sub-interface, existing ones are untouched:
Router(config)# interface gig0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
```

Because each VLAN has its own independent sub-interface, adding Gig0/0.40 for Spotify has no effect on the existing Gig0/0.10/.20/.30 sub-interfaces — Insta, Zomato and Flipkart continue to communicate exactly as before.

[Paste Packet Tracer screenshot here — Task 5 — show vlan brief and show ip interface brief confirming all four VLANs working together]

Layer 3 Switch Inter-VLAN Routing

Theory: Using a Layer 3 switch — no need for a router; uses SVIs (Switch Virtual Interfaces) for routing; faster than Router-on-a-Stick. Practical: Configuring inter-VLAN routing on a Layer 3 switch — enable IP routing.

1. Create three VLANs (Sales, Support, Marketing) on a Layer 3 switch

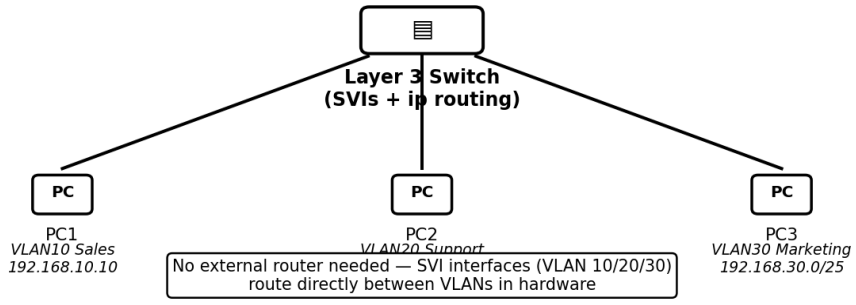


Fig: Inter-VLAN routing using SVIs on a Layer 3 switch (illustrative)

```
L3Switch(config)# vlan 10
L3Switch(config-vlan)# name Sales
L3Switch(config)# vlan 20
L3Switch(config-vlan)# name Support
L3Switch(config)# vlan 30
L3Switch(config-vlan)# name Marketing
L3Switch(config)# interface fa0/1
L3Switch(config-if)# switchport mode access
L3Switch(config-if)# switchport access vlan 10
```

[Paste Packet Tracer screenshot here — Task 1 — show vlan brief confirming VLAN 10/20/30 and their assigned interfaces]

2. Configure SVIs for each VLAN

```
L3Switch(config)# interface vlan 10
L3Switch(config-if)# ip address 192.168.10.1 255.255.255.0
L3Switch(config-if)# no shutdown
L3Switch(config)# interface vlan 20
L3Switch(config-if)# ip address 192.168.20.1 255.255.255.0
L3Switch(config-if)# no shutdown
L3Switch(config)# interface vlan 30
L3Switch(config-if)# ip address 192.168.30.1 255.255.255.0
L3Switch(config-if)# no shutdown
```

[Paste Packet Tracer screenshot here — Task 2 — show ip interface brief confirming all three SVIs are up/up]

3. Enable IP routing on the Layer 3 switch

```
L3Switch(config)# ip routing
```

Without this single global command, a Layer 3 switch will still create the SVIs and let each VLAN's own devices talk to their SVI gateway, but it will not route traffic between different VLANs — 'ip routing' is what turns on the actual routing function between the SVIs.

4. Verify inter-VLAN routing: ping between a VLAN 10 PC and a VLAN 20 PC

```
PC1> ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Reply from 192.168.20.10: bytes=32 time=1ms TTL=254
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Ping statistics: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

[Paste Packet Tracer screenshot here — Task 4 — captured ping results between VLAN 10 and VLAN 20 PCs]

5. Refactor VLAN 30 to use a /25 subnet mask instead of /24

```
L3Switch(config)# interface vlan 30
L3Switch(config-if)# ip address 192.168.30.1 255.255.255.128
```

Changing VLAN 30's SVI to a /25 mask (255.255.255.128) halves its address space to 126 usable hosts (192.168.30.1–192.168.30.126), and any device on that VLAN using an address in the upper half (192.168.30.128 onward) would now be on a different subnet and unable to reach the SVI gateway directly. This change is entirely local to VLAN 30 — it does not affect VLAN 10 or VLAN 20's own /24 addressing, and inter-VLAN routing between all three continues to work as long as every device within VLAN 30 is renumbered to fit inside the new, smaller /25 range.

a — Introduction to WAN Technologies: WAN Connections, CSU/DSU, WAN Aggregators

Theory: Definition of WAN, types of WAN connections, WAN devices: CSU/DSU, WAN aggregators.

A WAN (Wide Area Network) connects separate LANs over large geographic distances (cities, countries) using links leased from service providers.

1. Comparison of WAN connection types

Type	Speed	Cost	Typical use cases
Leased line (e.g. T1/E1)	Fixed, dedicated: 1.5–2 Mbps per T1/E1, higher with fibre leased lines	High — you pay for the dedicated circuit whether used or not	Head office to data center, banks, links needing guaranteed bandwidth
DSL	Moderate: roughly 1–100 Mbps depending on DSL type and distance	Low	Small branch offices, backup links, home/remote workers
MPLS	High and scalable (Mbps to Gbps), with QoS	Medium to high	Multi-branch companies needing reliable, prioritised traffic (voice, video, ERP)

2. CSU/DSU connecting a router to a WAN service provider

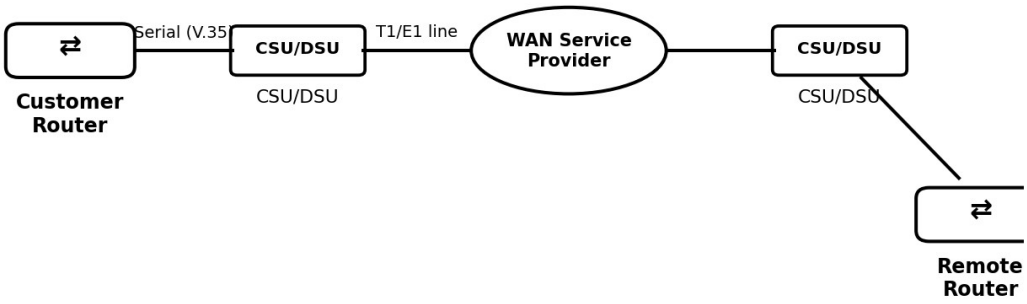


Fig: Router connected to a WAN service provider through a CSU/DSU (illustrative)

Router: the customer's router sends data from the LAN toward the WAN and chooses where packets go. CSU/DSU: it converts the router's serial (DTE) signals into the line format the provider's digital circuit (T1/E1) uses, provides clocking, and the CSU part protects and tests the line. WAN service provider: carries the traffic across its network to the remote site's CSU/DSU and router.

3. Three real-world apps that rely on WAN connections

- IRCTC: your booking request travels over the WAN/internet from your city to IRCTC's data centers, and payment gateways/bank networks are also reached over WAN links, so tickets can be booked from anywhere.
- WhatsApp: messages and media are sent over WAN links between your phone, WhatsApp's servers in remote data centers, and the receiver's phone, which is why chatting works across cities and countries.
- Zomato: the app in your hand, the restaurant's order screen and the delivery partner's app all connect to Zomato's cloud servers across WAN links, keeping orders and live tracking in sync.

4. WAN aggregator device (image/datasheet and key features)

Example device: Cisco ASR 1000 Series Aggregation Services Router (search this model name on Cisco's website to get its image and datasheet link).

- Key feature 1: high throughput and many WAN links - it aggregates traffic from many branch/WAN connections into one device, handling large volumes in hardware.
- Key feature 2: redundancy and scalability - supports redundant processors/power and modular interfaces, so a large enterprise or ISP edge can keep running and grow without replacing the device.

[Paste screenshot/image here — Task 4 — image of the device or the datasheet link you found]

b — WAN Authentication Protocols (HDLC, PPP, PAP, CHAP, MS-CHAP)

Theory: Why authentication is needed in WAN. WAN authentication protocols: HDLC, PPP authentication (PAP, CHAP, MS-CHAP).

1. Network diagram: where authentication is needed

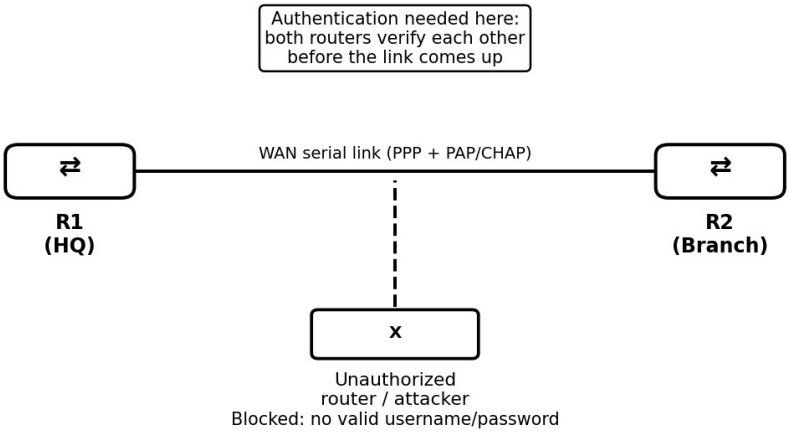


Fig: Authentication on a WAN link between two routers (illustrative)

Authentication is needed on the WAN link so each router proves who it is before the link is accepted; otherwise an unauthorized router or attacker could connect to the WAN link and receive or inject traffic.

2. HDLC WAN connection (and why it has no authentication)

```
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# encapsulation hdlc
R1(config-if)# clock rate 64000     ! DCE side only
R1(config-if)# no shutdown

R2(config)# interface serial 0/0/0
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# encapsulation hdlc
R2(config-if)# no shutdown
```

The link comes up and pings work, but no username or password was ever asked for. HDLC (Cisco's version is proprietary) only frames the data; it has no authentication feature, so any router configured with HDLC and a matching address can join the link. Missing in terms of security: identity checking.

[Paste screenshot/image here — Task 2 — show interfaces serial 0/0/0 (Encapsulation HDLC) and successful ping]

3. PPP with PAP authentication

```
R1(config)# username R2 password cisco
R1(config)# interface serial 0/0/0
R1(config-if)# encapsulation ppp
R1(config-if)# ppp authentication pap
R1(config-if)# ppp pap sent-username R1 password cisco

R2(config)# username R1 password cisco
R2(config)# interface serial 0/0/0
R2(config-if)# encapsulation ppp
R2(config-if)# ppp authentication pap
R2(config-if)# ppp pap sent-username R2 password cisco
```

If the password is incorrect on either side, PPP authentication fails: the interface stays 'up' but the line protocol goes 'down', pings fail, and debug ppp authentication shows a PAP authentication failure message. The link only comes up again once the passwords match.

[Paste screenshot/image here — Task 3 — show interfaces serial 0/0/0 with PPP/PAP, and the failed-password result]

4. Changing to CHAP

```
R1(config)# interface serial 0/0/0
R1(config-if)# ppp authentication chap

R2(config)# interface serial 0/0/0
R2(config-if)# ppp authentication chap

! username R2 / username R1 entries use the same shared password on both routers
```

Key difference: PAP sends the username and password in clear text, only once, when the link starts. CHAP never sends the password; the router sends a random challenge, the other side replies with a hashed (MD5) response, and the challenge is repeated periodically, so a captured packet cannot simply be replayed.

[Paste screenshot/image here — Task 4 — show running-config / show interfaces showing CHAP]

5. Real-world scenarios using these protocols

1. Mobile data/broadband login: when a phone or home router connects to an ISP over PPP/PPPoE, the ISP uses PAP/CHAP to check the subscriber's credentials. This is critical so only paying customers use the service and usage is billed to the right account.
2. Public Wi-Fi login (e.g. airport): the login page credentials are checked against an authentication server (often with CHAP/MS-CHAP style exchanges inside the Wi-Fi/RADIUS process). Authentication is critical so unknown people cannot join and misuse the network or the other users' traffic.

c — AAA (Authentication, Authorization, Accounting)

Theory: Three core functions, AAA in networking and security, AAA deployment models.

1. Scenario: user connecting to a private campus/airport Wi-Fi

Authentication: the user selects the Wi-Fi and enters a student ID and password. The access point passes them to the AAA server, which checks them against its database; wrong credentials = no access.
Authorization: once identity is proven, the server decides what the user may do — e.g. a student gets internet and library resources but not the admin network, and may be limited to certain sites or hours.
Accounting: while connected, the system records when the user logged in, how long they stayed, the device and the amount of data used. This log is used for security audits, troubleshooting and (at airports) usage limits or billing.

2. AAA deployment models (Zomato-type company)

	Centralized AAA (dedicated server, e.g. RADIUS/TACACS+)	**Local / distributed AAA (users stored on each device)**
Advantages	1. One place to add/remove users and change policy 2. Consistent rules and complete central logs for audits 3. Easily scales to many devices and offices	1. No extra server needed, simple for small setups 2. Works even if the network to the server fails 3. Low cost and quick to configure
Disadvantages	1. Server is a single point of failure unless made redundant 2. Needs extra hardware, setup and maintenance 3. Server holds sensitive data so becomes a target and must be heavily protected	1. User accounts must be changed manually on every device 2. Inconsistent policies and hard to audit 3. Does not scale; poor for a company holding sensitive customer data

3. Network diagram: AAA servers in an enterprise network

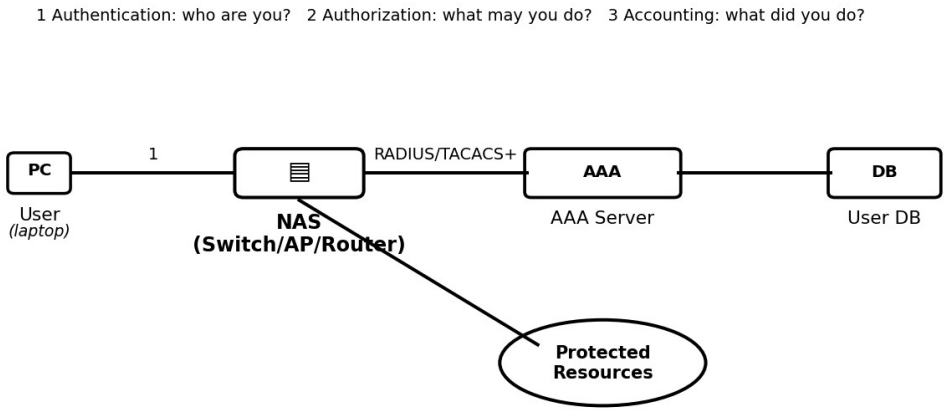


Fig: AAA server integrated into an enterprise network (illustrative)

Authentication happens when the user's credentials are sent from the NAS (switch/AP/router) to the AAA server and checked against the user database. Authorization happens next, when the server replies with what the user may access. Accounting happens throughout the session, as the NAS sends usage records to the server.

4. AAA applied to an app (Paytm)

When a user logs into Paytm, authentication confirms identity with the mobile number, password/PIN and OTP or biometric. Authorization then decides what that verified user can do — for example, a basic account may have lower transaction limits than a full-KYC account, and only the account holder can send money from their wallet. Accounting records every action: login time and device, each payment, and each failed attempt. These records support fraud detection, dispute resolution, statements and regulatory audits, much like the tracking used for security or billing.

5. Sample AAA policy for a college Wi-Fi (AI-generated sample)

This sample policy was drafted with an AI tool for the assignment; review it before submitting.

```
COLLEGE Wi-Fi AAA POLICY
1. Authentication: Only students with a valid college ID and password
   (verified against the college directory via a RADIUS server) may connect.
   Guests and unknown devices are denied.
2. Authorization: Students may access the internet and educational sites;
   social-media, streaming and gaming sites are blocked during class hours.
   Staff-only systems are not reachable from the student SSID.
3. Accounting: Every login, logout, device MAC, IP address and data usage is
   logged. Failed attempts are logged and reviewed weekly; logs kept 12 months.
```

Authentication is addressed by the ID/password check and denial of unknown users; authorization by the allowed/blocked website rules and restricted network areas; accounting by the logging of all access attempts and usage.

d — RADIUS & TACACS+

Theory: TACACS+ and RADIUS features, RADIUS authentication process, TACACS+ use cases, RADIUS vs. TACACS+.

1. RADIUS vs TACACS+ comparison

Feature	RADIUS	TACACS+
Protocol type	Open standard (IETF); uses UDP	Cisco-developed (now widely supported); uses TCP
Encryption	Encrypts only the password in the packet	Encrypts the entire packet body
Ports used	UDP 1812/1813 (older 1645/1646)	TCP 49
Command authorization	Limited; combines authentication and authorization, not per-command	Separates AAA functions; supports per-command authorization for admins
Typical use	Network access for users (Wi-Fi, VPN, dial-in)	Administrator access to network devices

2. RADIUS authentication sequence for a Wi-Fi login

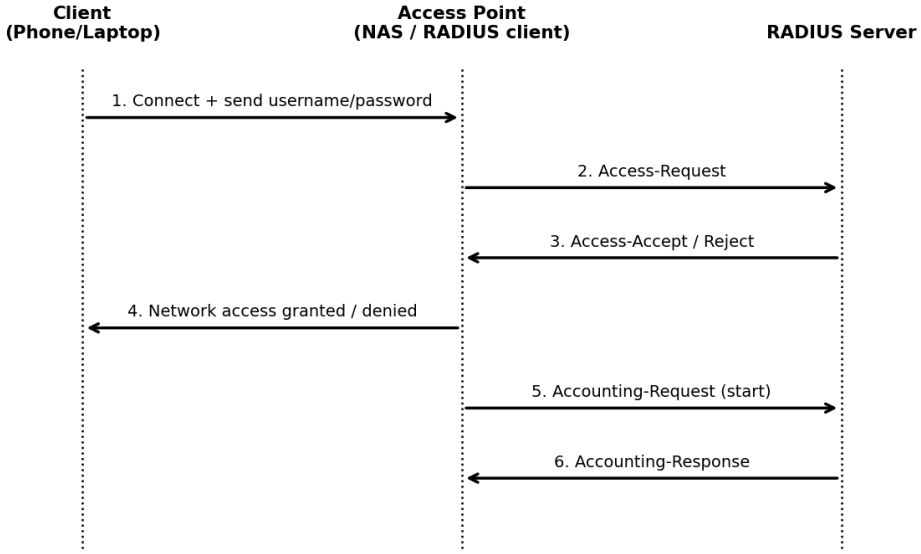


Fig: RADIUS authentication sequence for a Wi-Fi login (illustrative)

The user connects and enters credentials; the access point (RADIUS client) sends an Access-Request to the RADIUS server; the server checks the user database and returns Access-Accept or Access-Reject; the AP then grants or blocks the user and sends accounting messages while the session lasts.

3. Real-world scenario where TACACS+ is more suitable

In a company like IRCTC, the network team has many engineers logging in to routers and firewalls that carry booking traffic. TACACS+ is more suitable because it can authorize each command: a junior engineer may be allowed only 'show' commands while a senior engineer may change routing configuration. It also encrypts the whole session and logs every command an admin typed, so changes can be audited. RADIUS cannot control admin access with this level of detail.

4. RADIUS in a public hotspot (malls/airports)

Advantages:

- Centralized, scalable: one RADIUS server can authenticate users across many hotspot access points and locations.
- Open standard supported by almost all Wi-Fi equipment, with accounting data for usage limits and billing.

Disadvantages:

- Only the password is encrypted, so other information in the packet is exposed unless the link is protected (e.g. IPsec/RadSec).
- Uses UDP, which has no built-in delivery guarantee, and the server is a single point of failure unless made redundant.

5. Sample TACACS+ configuration for a Cisco router (AI-generated sample)

Sample generated with an AI tool using the prompt 'basic TACACS+ configuration for Cisco router'; lines to modify for a college lab are marked.

```
aaa new-model                                    ! enables AAA (needed first)
tacacs-server host 192.168.1.100 key LabKey123  ! CHANGE: lab server IP and key
aaa authentication login default group tacacs+ local
aaa authorization exec default group tacacs+ local
aaa accounting exec default start-stop group tacacs+
username labadmin privilege 15 secret LabAdmin!  ! CHANGE: local fallback account
line vty 0 4
  login authentication default
```

Lines to modify for a college lab: the tacacs-server host IP address and shared key must match the lab's TACACS+ server, and the local fallback username/password should be changed. The 'local' keyword is kept so staff can still log in if the server is unreachable. 'aaa new-model' stays as it is, since it turns on AAA.

e — Hybrid WAN (SD-WAN)

Theory: Introduction to SD-WAN, key benefits of SD-WAN.

1. Traditional WAN vs SD-WAN

Aspect	Traditional WAN	SD-WAN
How data is routed	Fixed paths, mostly over private links like MPLS, set by static routing	Dynamic path choice per application across MPLS, broadband and 4G/5G based on live link quality
Management	Each router configured separately at each site, often through CLI	Centralized controller/dashboard pushes policy to all sites
Security	Traffic usually backhauled to head office; security added by separate devices	Built-in encryption and segmentation, with security applied at each branch
Cost	Expensive private circuits and long provisioning times	Uses cheaper broadband alongside MPLS; faster, often zero-touch setup
Cloud access	Cloud traffic goes through the data center, adding delay	Direct local internet breakout to cloud apps like Office 365/AWS

2. SD-WAN diagram

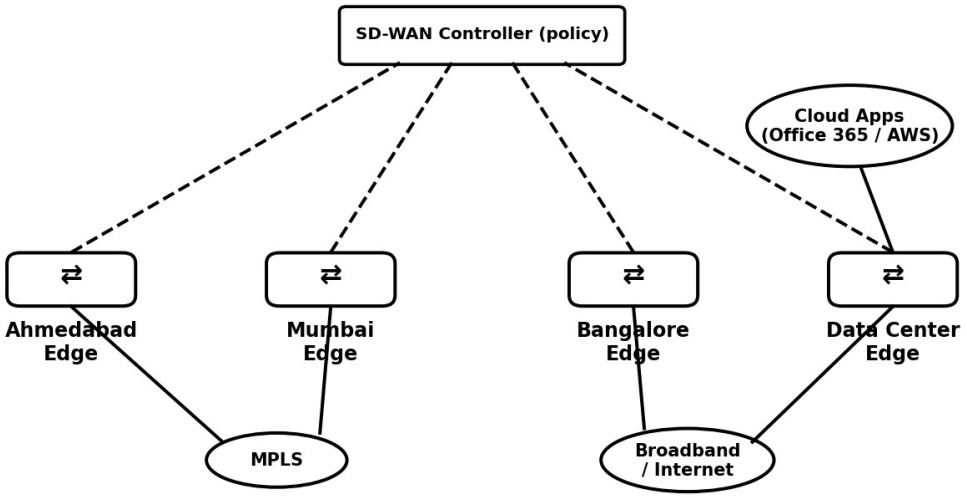


Fig: SD-WAN connecting branches, data center and cloud apps over MPLS and broadband (illustrative)

The SD-WAN controller (top) manages policy for all edge devices; each branch edge connects to the main data center and cloud apps over two WAN links: MPLS (reliable, private) and broadband/Internet (cheap, high bandwidth).

3. Scenario: Zomato across multiple cities

Imagine Zomato's offices and support centers in Ahmedabad, Mumbai and Bangalore all use Zomato's order and tracking systems at the same time. With SD-WAN, the order and live-tracking traffic is marked as high priority and automatically sent over the best-performing link, and if one link slows down or fails, traffic moves to the other within moments. Users and staff see faster, more reliable order updates, and security policies are applied the same way in every city.

4. Three key benefits of SD-WAN (Ahmedabad, Mumbai, Bangalore offices)

1. Lower cost: broadband can carry part of the traffic, reducing expensive MPLS usage. Example: the Ahmedabad office routes routine web and email over broadband and keeps MPLS for critical systems.
2. Better reliability and performance: it monitors links and moves traffic when one degrades. Example: if the Mumbai broadband link drops during a video call with Bangalore, the call shifts to MPLS or another link without disconnecting.

3. Simple central management and security: one dashboard applies policies to every site. Example: IT can set up a new Bangalore branch in a day by shipping a pre-configured edge device, without sending an engineer from Ahmedabad.

f — AAA (Authentication, Authorization, Accounting)

Theory: Three core functions, AAA in networking and security, AAA deployment models.

1. Comparison table: Authentication vs Authorization vs Accounting

Function	What it answers	Real-world examples
Authentication	Who are you?	1. Instagram: logging in with password plus a 2-factor code. 2. Paytm: logging in with a PIN, OTP or fingerprint. 3. Zomato: logging in with a mobile number and OTP.
Authorization	What are you allowed to do?	1. Instagram: you can edit only your own posts and profile; business accounts also get insights. 2. Paytm: transaction limits depend on whether your KYC is complete. 3. Zomato: a restaurant partner can edit only its own menu; a delivery partner sees only assigned orders.
Accounting	What did you do, and when?	1. Instagram: the list of devices/locations where your account is logged in. 2. Paytm: transaction history and statements. 3. Zomato: order history and delivery logs used for billing and disputes.

2. Network diagram: AAA securing a college campus Wi-Fi

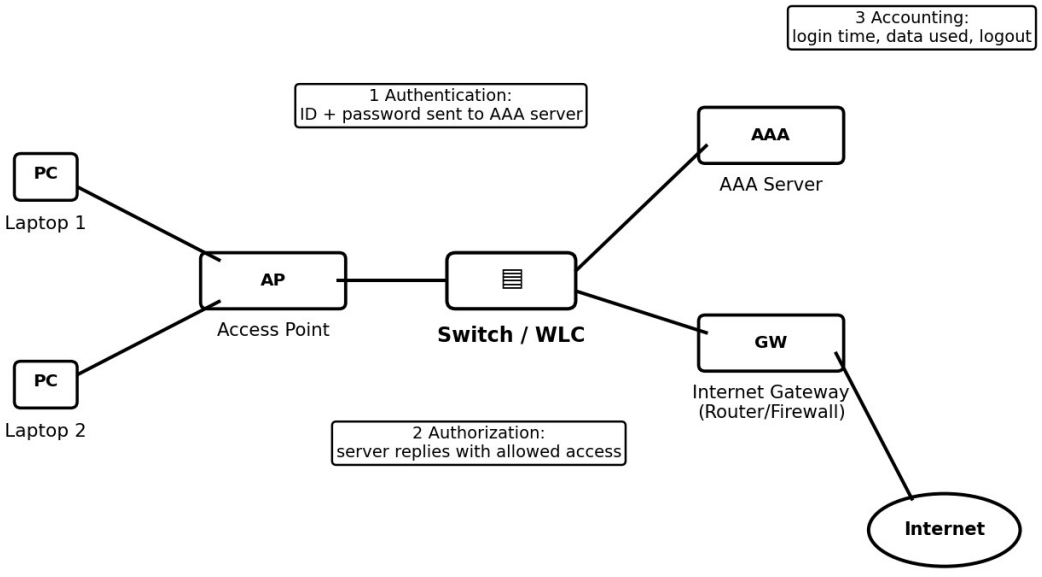


Fig: AAA securing access to a college campus Wi-Fi network (illustrative)

Authentication: the laptop's ID and password pass through the access point and switch to the AAA server, which checks them. Authorization: the AAA server replies with what the student may access (for example internet and library only). Accounting: the network device sends records of login time, data used and logout to the AAA server. Once allowed, the traffic goes through the internet gateway to the internet.

3. Centralized vs distributed AAA for multiple Flipkart warehouses

Centralized model: all warehouses send their login requests to one AAA server (or a redundant pair) at the head office. Staff accounts and rules are created once, so when an employee leaves, one change removes access to every warehouse, and all logs are in one place. If the link to the head office fails, logins at the warehouses can fail unless there is a backup.

Distributed model: each warehouse has its own AAA server (or stores users on its own devices). Logins keep working even if the head-office link is down, but accounts and policies must be managed separately at each warehouse, so they can become inconsistent, and the logs are spread across many places.

4. Centralized AAA for a large-scale app like BookMyShow

Advantages:

- One place to manage users and roles, so policy changes (for example, for staff at many theatres) are applied everywhere at once.
- Complete, central logs make it easier to detect fraud such as ticket-blocking abuse and to prepare audits.
- Scales well: new servers, offices or theatres simply point to the same AAA service.

Disadvantages:

- The server is a single point of failure during peak sales unless it is made redundant, which adds cost.
- It holds all credentials in one place, making it an attractive target that must be heavily secured.

5. Sample AAA policy for a Zomato-like app (AI-generated sample)

This sample policy was drafted with an AI tool for the assignment; the highlighted line is the one to change.

```
ZOMATO-LIKE APP: AAA POLICY
AUTHENTICATED ACTIONS
  - Customer: login (OTP), place order, change address, make payment
  - Restaurant partner: login (password), edit menu, view payouts
  - Delivery partner: login (OTP), accept order, update delivery status
  - Admin: login (password only)                    <<< CHANGE THIS
AUTHORIZED ROLES
  - Customer: own profile, own orders only
  - Restaurant partner: own restaurant menu and orders only
  - Delivery partner: only orders assigned to them
  - Admin: user management, refunds, system settings
ACCOUNTING LOGS
  - Logins (time, device, location), failed attempts
  - Orders and payments (who, what, amount, time)
  - All admin actions (who changed what, old and new value)
  - Logs retained for 12 months
```

One thing I would change to make it more secure: admins are authenticated with a password only. Admins can issue refunds and change system settings, so I would require multi-factor authentication (password plus an authenticator app or hardware key) for every admin login. A stolen password alone then cannot give an attacker admin access.

g — RADIUS & TACACS+

Theory: TACACS+ and RADIUS features, RADIUS authentication process, TACACS+ use cases, RADIUS vs. TACACS+.

1. Side-by-side feature comparison

Feature	RADIUS	TACACS+	Which protocol supports it
Open standard (IETF), works across most vendors	Yes	Originally Cisco-developed	RADIUS (TACACS+ is mainly Cisco-oriented, though others support it)
Encrypts the entire packet body	No - only the password	Yes	TACACS+
Per-command authorization for admins	Limited	Yes	TACACS+
Separates authentication, authorization and accounting	No - combines authentication and authorization	Yes - separate	TACACS+
Transport and ports	UDP 1812/1813	TCP 49	Both, with different transport
Network access (Wi-Fi 802.1X, VPN)	Yes - the standard choice	Not typical	RADIUS

2. Flow diagram: RADIUS authentication on a public Wi-Fi hotspot

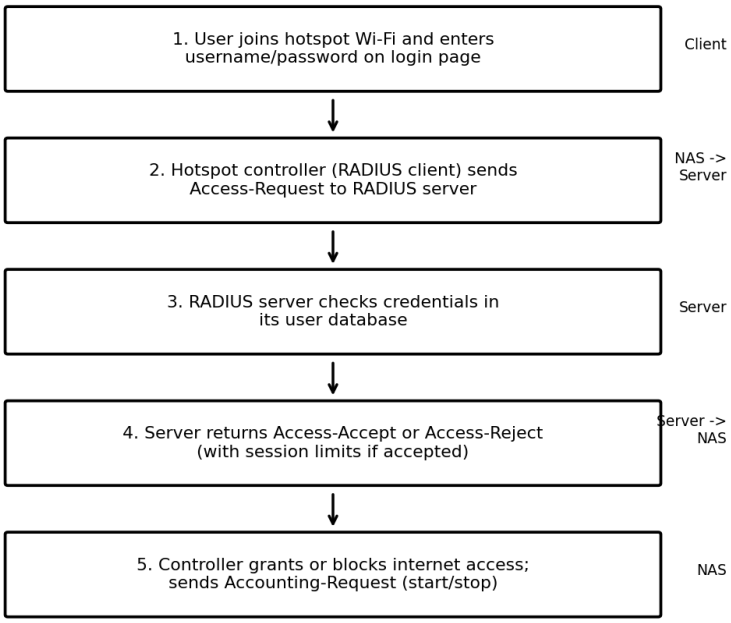


Fig: RADIUS authentication flow for a public Wi-Fi hotspot (illustrative)

Client to network device: the user enters credentials. Network device to RADIUS server: Access-Request. RADIUS server to network device: Access-Accept or Access-Reject. Network device to client: access granted or blocked, followed by accounting messages to the server.

3. Three use cases where TACACS+ is preferred over RADIUS

1. Administrator access to core routers and switches: TACACS+ can authorize each command, so a junior engineer is limited to 'show' commands while a senior engineer can change configuration. RADIUS cannot control commands at this level.

2. Large data centers with several teams: separate command sets can be given per role (network team, security team, auditors) from one server, because TACACS+ keeps authentication, authorization and accounting separate.
3. Banks and payment companies (like a Paytm-type environment) with strict audits: TACACS+ encrypts the whole session and records every command an admin typed, giving a full trail of who changed what.

4. A failed login to a network device: how each protocol responds

	RADIUS	**TACACS+**
Response	Server returns Access-Reject to the network device.	Server returns a FAIL/REJECT reply for the authentication step.
Logging of the failure	The failed attempt is recorded in the server's authentication log. Accounting normally covers sessions, so a failed login that never starts a session usually has no accounting record.	The failed attempt is logged with user, source device and time, in the same server that handles authorization and accounting.
Command authorization	Not applicable at this stage; RADIUS has no per-command control, so even after a successful login it cannot approve or deny individual commands.	No commands are authorized because login failed; after a successful login, every command could be authorized and its accounting record logged.

5. Sample RADIUS configuration for a Cisco router (AI-generated sample)

Sample generated with an AI tool; changes for a college campus Wi-Fi setup are listed after it.

```
aaa new-model
radius server COLLEGE-RADIUS
  address ipv4 192.168.1.100 auth-port 1812 acct-port 1813
  key LabKey123
aaa authentication login default group radius local
aaa authorization network default group radius
aaa accounting network default start-stop group radius
```

Changes I would make for a college campus Wi-Fi:

- Replace the server address with the campus RADIUS server, and use a long, random shared key instead of the sample key.
- Use 802.1X for Wi-Fi: add 'aaa authentication dot1x default group radius' and 'dot1x system-auth-control', and use WPA2-Enterprise on the SSID so each student logs in with their own ID.
- Add a second RADIUS server (backup) so students are not locked out if one server fails, and keep 'local' as a fallback only for the admin account.
- Keep accounting on so login times and data usage are logged for security reviews.